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Automated Invigilation System for Offline Exam Monitoring using Deep Learning

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***Abstract*—Exam proctoring is an exhausting job since it gets more diligently for the supervisors to keep an eye on the students in the testing area. In the field of education, determining suspicious conduct during exams is crucial because it tackles the vital problem of upholding academic integrity. It benefits both educational institutions and students by fostering an environment that is fair and deserving of trust for assessments. The majority of the current methods depends on traditional invigilation systems, which are difficult to use and mainly rely on human supervisors. They not only require a large amount of labor, which raises costs, but also a great deal of coordination and training. In this work, a more dependable, effective, and technologically advanced method for tracking and evaluating activities during offline examination is presented. The proposed system uses the internet of things (IOT) combined with sophisticated deep learning techniques to identify and detect suspicious activity and noisy hot-spots. For training and evaluation, we utilized benchmark dataset, xamDataSetv2, for our experiments and achieved 98%, 73%, 91%, and 80.6% of accuracy, precision, recall, and f1-score respectively.**

***Keywords*—deep learning, offline examination, cheating, detection, image processing.**

I. Introduction

In today's dynamic educational landscape, examination integrity is a critical component of academic credibility [1]. With technological advancements, traditional offline examination methods are being revitalized through innovative solution, one of which is the Automated Invigilation System for Offline Exam Monitoring based on Deep Learning Adil [2]. This system represents a paradigm shift in exam invigilation, taking advantage of Deep Learning, a subset of artificial intelligence, to improve monitoring processes [3]. As educational institutions adopt modern assessment methodologies, the need for strong invigilation mechanisms becomes more pressing.

Conventional invigilation techniques frequently require significant resource allocation and are prone to human error. However, the Automated Invigilation System overcomes these challenges by incorporating Deep Learning algorithms, which automate the monitoring process[29]. The system can detect and flag suspicious activities such as cheating based on video feed analysis [4]. Furthermore, the use of object detection technology ensures student authentication and reduces the risk of identifying fraud [5].The purpose of this report is to investigate the Automated Invigilation System's transformative potential, including its architectural complexities, functional mechanisms, and potential implications for examination security. This study aims to provide a comprehensive understanding of the system's role in shaping the future of invigilation practices for educational institutions around the world [6].

A. Motivation and Contribution

Traditional exam proctoring presents significant challenges due to the difficulty of monitoring students in exam halls [7]. Relying on manual supervision necessitates a large workforce, increasing costs for educational institutions. Furthermore, extensive training and coordination among invigilators is required to ensure that exams run smoothly and irregularities are detected. The labour-intensive nature of traditional proctoring frequently causes stress and pressure on supervisors as they strive to maintain academic integrity and prevent cheating behaviours [8].

However, technological advancements present promising alternatives. Automated proctoring systems use deep learning and machine learning algorithms to remotely monitor students via audio, video, and screen capture analysis [9]. These systems offer real-time monitoring and flag suspicious behaviour for further investigation. Automated proctoring solutions reduce reliance on manpower and streamline the monitoring process, providing cost-effective and efficient alternatives that improve assessment integrity while alleviating the burdens associated with traditional exam supervision. The recommended strategy improves academic integrity, minimizes the need for staff, and offers a reliable setting for assessment [10]. It has significance because it will transform exam monitoring, provide fairness, and advance the creation of technologically advanced educational solutions

II. Related Work

The literature on Online Exam Proctoring Systems (OEPS) based on Artificial Intelligence (AI) developed a response to the growing need for secure assessment methods brought about by the rise of online education by addressing the short comings of conventional techniques [11]. These technologies help to create a more secure online testing environment. These days, biometrics and behavioural analysis are crucial parts of AI- based OEPS. The use of AI to examine user behaviour on tests, such as mouse clicks, keystroke patterns, and gaze direction. The integrity of online assessments is strengthened by these studies, which aid in the development of proctoring systems that can identify irregularities suggestive of cheating or impersonation. Another important area of research is machine learning applications for cheating detection to analyse interaction patterns, spot odd behaviours, and provide real-time alerts. The need for ongoing observation and flexible reactions to changing cheating tactics is addressed by these developments.

Deep learning, a powerful subset of machine learning, has significantly advanced various scientific and technological fields by enabling systems to learn complex patterns [13, 32]. Various power deep learning algorithms such as Convolutional Neural networks [18], Recurrent Neural Networks [32], and Attention models [34] have been utilized in image, text, and sequence data. The researchers are working to create advanced systems that can identify suspicious behaviour that could be a sign of cheating because it is crucial to up-hold academic integrity during exams [12]. Initial research provided the foundation for recognizing and comprehending different types of exam cheating practices. Recently, for example, advances have been made in real-time activity recognition through the integration of machine learning and computer vision techniques. The process of keeping track of a behaviour, activity, or unusual behaviour with the aim of regulating, managing, protecting, or influencing it is commonly referred to as surveillance [33]. Surveillance is used by government agencies to find out about crimes, obtain intelligence, or protect a process, a person, or a point [22]. On the use of video surveillance in examination rooms for automated detection of cheating, automated cheating detection systems are being investigated as a result of the changing educational landscape and the need for strong mechanisms to maintain academic integrity[25]. Early studies demonstrated the short comings of conventional methods of monitoring in exam rooms. This led to a move towards automated video surveillance systems that made use of developments in machine learning and computer vision.

III. Proposed Method

The YOLOv8 algorithm as shown in Fig. 1 has four major components: input, backbone, neck, and output. It is a quick, one-step object detection method and consists of following steps.

A. Initialize Parameters

Specify the height (H), width (W), and number of channels (C) of the input images. This shape is crucial as it determines how the data will flow through the network. The number of classes are set to 3 in our work that the model will predict.

B. Backbone Network

The backbone is responsible for processing the input image and extracting useful spatial hierarchies of features. This is achieved through successive convolutional layers. Conv2D layers are used for the convolution operations that capture the spatial features in the images, while MaxPooling2D layers are applied to reduce the spatial dimensions of the feature maps, which help in reducing the computational load and overfitting.

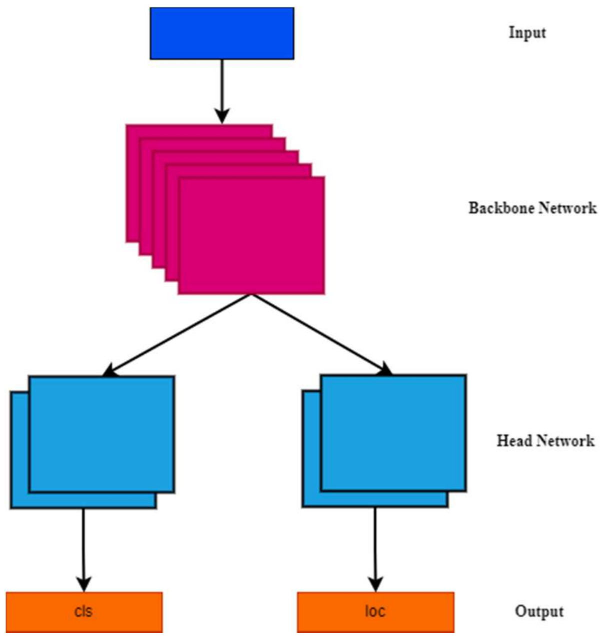

Fig. 1. Sample frames of classroom exam invigilation dataset.

C. Neck

This part of the network takes the feature maps from the backbone and further refines them, enhancing the ability to detect features at various scales. Upsampling is used to increase the resolution of the feature maps, allowing the network to detect objects at different scales more effectively. This is crucial for handling different object sizes within the same image.

D. Head

The head of the network is responsible for making the final predictions. It uses additional Conv2D layers to predict bounding boxes around detected objects, objectness scores to measure the presence of objects within the boxes, and class probabilities to classify the detected objects. The final output is usually a tensor that combines the bounding box coordinates, objectness scores, and class probabilities for each detected object.

E. Combine Components

The three components (Backbone, Neck, and Head) are combined into a single model. The backbone feeds into the neck, which in turn feeds into the head, creating a seamless flow for data processing and prediction. The model is compiled using an optimizer like Adam, which helps in minimizing the loss during training. The loss function is typically a variant of YOLO loss, which is tailored to handle the complexities of bounding box prediction along with classification.

The input segment improves images through grayscale padding, adaptive anchor calculation, and mosaic augmentation. To provide multi-scale feature maps, the backbone and neck of YOLOv8 were extracted using the Conv and C2f modules. The C2f module, an improved version of the C3 module, uses the YOLOv7 ELAN structure to reduce the need for a convolutional layer and improve gradient flow through the Bottleneck module. Before proceeding to the neck, the SPPF module combines feature maps with various kernel sizes. The neck's FPN+PAN structure improves multi-scale detection by combining high- and low-level features via up sampling and down sampling. The detection head distinguishes between detection and classification by assigning samples based on the sum of the classification and regression scores via TaskAlignedAssigner. Binary Cross-Entropy is used for classification, Distribution Focal Loss, and CIoU for regression, and the loss calculation excludes the Objectness branch. With the help of a task alignment metric, YOLOv8's decoupled heads improve prediction by forecasting regression coordinates and classification scores [19]. By suppressing low-quality predictions, this metric maximizes detection accuracy by combining classification scores and IoU values. When an object's IoU value exceeds 0.5, it is considered detected. This metric evaluates detection accuracy.

The Improved YOLOv8 Model is to deploy YOLOv8 on embedded platforms, the model structure was changed to reduce parameters and computational complexity while maintaining accuracy. Using MobileNet v3 as the backbone and integrating LSK-attention and BiFPN improved accuracy while increasing complexity. Using the SimSPPF module mitigated this, resulting in a shorter inference time. In YOLOv8, the PAN-FPN structure improves semantic and localization features by combining B3-P3 and B4-P4, followed by P4-N4 and P5-N5[20]. However, it does not handle large-scale feature maps well and suffers from information loss during up sampling and down sampling, indicating that detection quality could be improved further. To improve YOLOv8, the Bi-directional Feature Pyramid Network (BiFPN) was added, which takes advantage of efficient bi-directional cross-scale connections and weighted feature fusion. This method, inspired by EfficientDet, improves semantic information and spatial accuracy, especially for small targets like road cracks. The model's receptive field expands as new paths are introduced from backbone feature maps, increasing detection accuracy [21]. To improve YOLOv8, the Bi-directional Feature Pyramid Network (BiFPN) was added, which takes advantage of efficient bi-directional cross-scale connections and weighted feature fusion. This method, inspired by EfficientDet, improves semantic information and spatial accuracy, especially for small targets like road cracks. The model's receptive field expands as new paths are introduced from backbone feature maps, increasing detection accuracy. To improve real-time road defect detection, YOLOv8 replaces the SPPF module with SimSPPF from YOLOv6 [27]. SimSPPF uses three 5x5 max pooling layers and ReLU activation to improve receptive field and feature representation while reducing computational complexity and processing time. In (1) and (2) show the exact formulas for these activation functions.

$$ReLU : f(x) \begin{cases} x, if\ x > 0 \\ 0, if\ x \leq 0 \end{cases} = max\ (0,\ x) \tag{1}$$

$$SiLU : f(x) = \frac{x^2}{1+e^{-x}} \tag{2}$$

In SimSPPF, substituting ReLU for SiLU reduces computational complexity, stops gradient vanishing, and speeds up convergence. The benefits of ReLU have been demonstrated by experimental comparisons, which reveal ConvBNReLU modules to operate 18% faster than ConvBNSiLU modules [30].

IV. RESULTS AND DISCUSSION

A. Datasets

In this paper, we used publicly available ExamDataSetv2 dataset which contains CCTV footage from the exam space [31]. The dataset contains 1260 images of classroom-based exams. The sample frames of classroom exam invigilation dataset extracted from CCTV camera are shown in Fig. 2.

Fig. 2. Sample frames of classroom exam invigilation dataset.

B. Dataset Annotation

The first step towards building a successful prediction method for detection of suspicious behavior is Data annotation, a process of adding labels to different objects in an image in order to make it understandable to machine learning algorithms [15]. The LabelImg [16], a free and open-source image annotation program, is used for annotating images with bounding boxes as shown in Fig. 3. The Preprocessing of CCTV footage involves several steps that must be completed in order to convert raw video data into a machine learning compatible format [17]. The video is first converted to a standardized format. After that, individual frames are extracted, and temporal down sampling may be used to reduce the frame rate [14]. We annotated all the objects into three categories: Person, Student Cheating, and Student Not Cheating [35].

Fig. 3. Annotation and Labelling of Images

C. Evaluation Metrics

1) Accuracy: Classification accuracy is the accuracy we generally mean, whenever we use the term accuracy and is computed using (3).

$$Accuracy = \frac{No.of\ correct\ predicitions}{Total\ no.of\ input\ samples} \quad (3)$$

2) Recall: Recall is to calculate the proportion of actual positive that was identified incorrectly and is computed using (4).

$$Recall = \frac{TP}{TP+FN} \quad (4)$$

3) Precision: Precision measures the percentage of predictions made by the model that are correct and is computed using (5).

$$Precision = \frac{TP}{TP+FP} \quad (5)$$

4) F-score: It is a harmonic mean between recall and precision and is computed using *(6).*

$$F - score = 2 * \frac{1}{\frac{1}{pre}+\frac{1}{recall}} \quad (6)$$

D. Experimental Setup

TensorFlow and PyTorch (version 2.1.0) were utilized in the Google Collab experiments to train neural networks. An Intel Xeon CPU, 100 GB of SSD capacity, 25 GB of RAM, and a Tesla K80 GPU were each used in the configuration. 30% of the 640x640 image data was set aside for testing, while the rest of 70%, was used for training. Early halting was used with a 50- epoch patience threshold to avoid over fitting [26]. Early stopping stops training if there is no improvement after 50 consecutive epochs of monitoring model performance on an evaluation set. The number of training examples handled in each iteration was set at a batch size of 16. The training process demonstrated over the whole dataset twenty-five times, or epochs [28]. Utilizing Google Colab's resources, this traditional deep learning configuration seeks to effectively train and assess neural networks.

E. Results

The results that were obtained after giving input videos into the system is shown in Fig. 4.

Fig. 4. Results of proposed model for cheating detection.

True positives, true negatives, false positives, and false negatives are highlighted in a confusion matrix shown in Fig. 5 for the proposed model, thus making more straight forward to conduct comprehensive performance analysis and evaluate predictive accuracy is computed from (1).

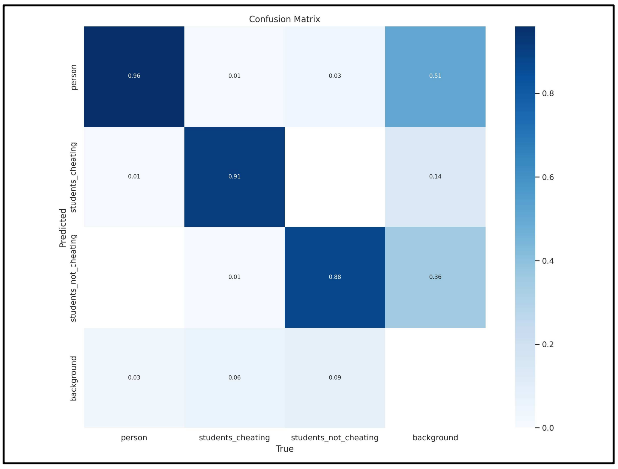

Fig. 5. Confusion Matrix of proposed model.

The results for proposed model, indicate metrics for training and validation, that include box loss, classification loss, and precision-recall values. The measured fluctuations show how effectively the model performs during training and assessment over epochs[23]. The data is extracted from the exam surveillance and video's at BVRIT Hyderabad

College of Engineering for women and annotated the extracted frames using LabelImg. We trained YOLOv8 model on the constructed dataset for detecting suspicious cheating behaviour. Our model achieved 73%, 91%, and 80.6% precision by calculating equation (2), recall computed equation (3), and f-score determined from equation (4). Our results are comparable with existing methods [24].

F. Performance Comparision

We compared the performance of our method with recent baseline approaches. As shown in table 1, our method achieved the accuracy of 98% with 73% precision, 91% recall and f-score which is 80.6%. In addition, we compared our accuracy to that of M D et al. and discovered that our approach exceeds the existing one.

TABLE I. PERFORMANCE COMPARISON OF THE PROPOSED MODEL WITH EXISTING METHODS

Authors	Method	Acc	Pr	Rc	F1
Satre et al.,2023 [11]	YOLO algorithm	85.3%	-	-	-
Trabelsi1 et al., 2021 [28]	YOLOv7	71.9%	-	-	-
Genemo et al., 2021[12]	Variation versions of SVM and KNN categorizers	92.99%	-	-	-
Baile et al.,2022[4]	Convolutional neural network-based machine learning model	92.9 %	-	-	-
Borse et al.,2020 [29]	TSC based and sRNN	74.6%	-	-	-
Proposed method	YOLOV8	98%	73%	91%	80.6%

V. CONCLUSION

Automated Invigilation System for Offline Exam Monitoring is a significant step toward maintaining academic integrity and fostering trust in educational assessment processes. This system enables educational institutions to ensure fair and transparent examinations while maintaining the value and credibility of academic credentials by leveraging the power of deep learning technology. In this project, a comprehensive solution was developed for Invigilation for Offline Exam. This has made an important breakthrough in improving the accuracy and robustness of examination monitoring with the integration of the YOLOv8 algorithm .This YOLOv8- enhanced Automated Invigilation System appears as a comprehensive solution to address the challenges raised by conventional invigilation strategies. It ensures a more secure and reliable exam environment by optimizing the monitoring process and adding a layer of adaptability to various exam scenarios. This blend of strategic enhancements and state of the art technology has great potential for the future of examination integrity in offline settings. The development of an Automated Invigilation System for Offline Exam Monitoring based on Deep Learning represents a significant step forward in ensuring the integrity and fairness of examination processes. Through the integration of deep learning techniques, such a system provides a sophisticated solution for detecting and discouraging cheating behaviours in offline exam environments. In future, the incorporation of noise modulation into an offline Automated Invigilation System represents a step forward in enhancing the exam environment. First of all, noise modulation serves as a discouragement for any attempt by examinees to interact with one another without permission in order to cheat. The system disrupts with low-volume conversations and other covert methods of communication by adding controlled ambient noise. By discouraging dishonest practices, this proactive measure helps to maintain the integrity of the examination setting. By adding noise modulation, the Automated Invigilation System grows an additional tool for maintaining academic integrity since it not only strengthens the system's security measures but also assures that it can be modified to various offline exam environments

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